



Department of Chemistry

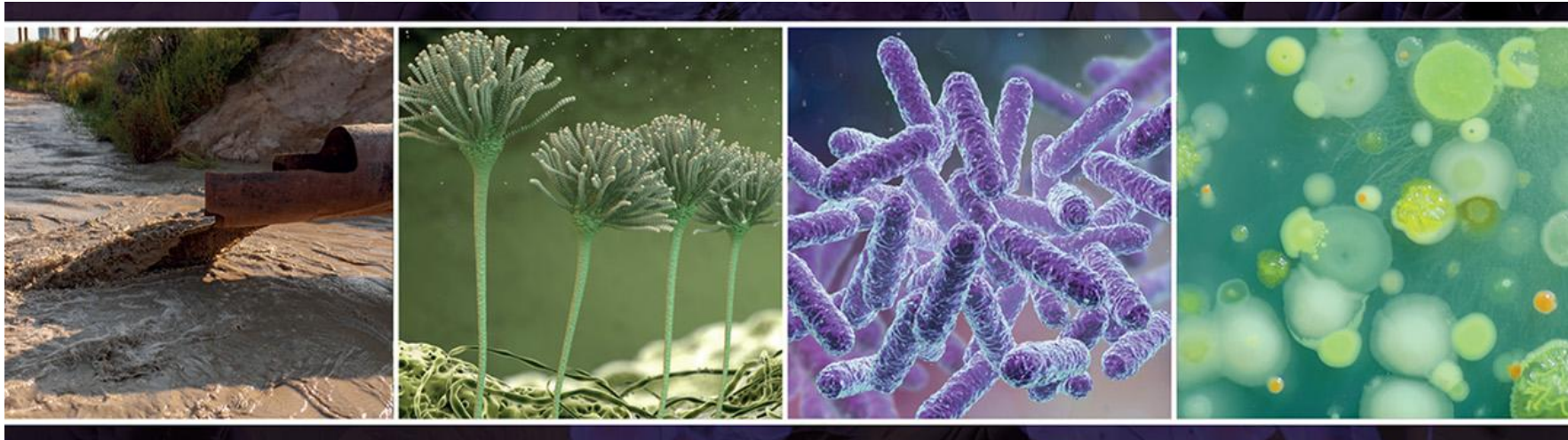
Environmental Chemistry (2 credit)

CY1010

July 2026

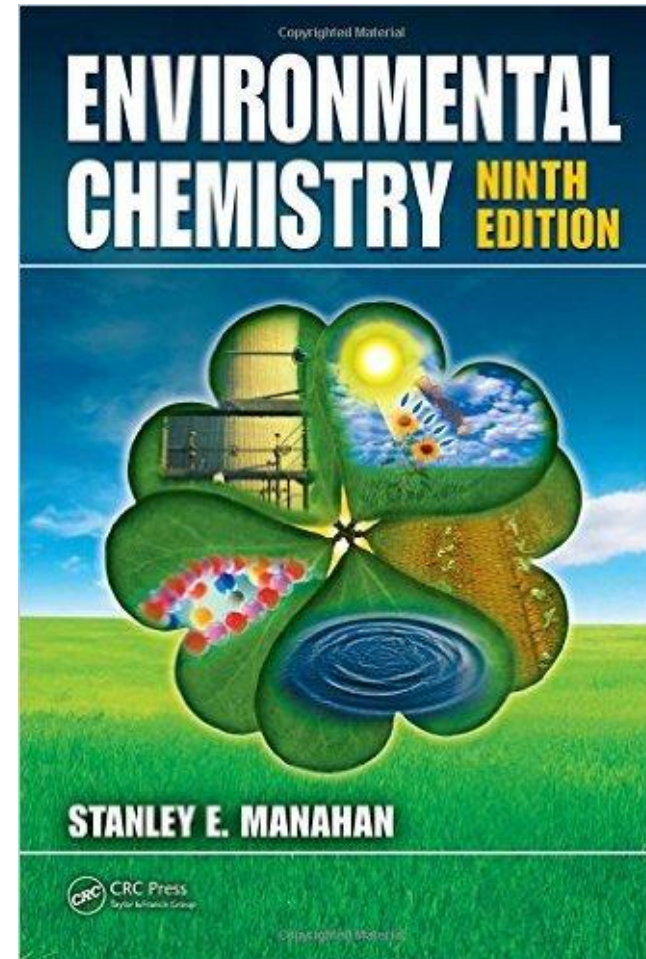
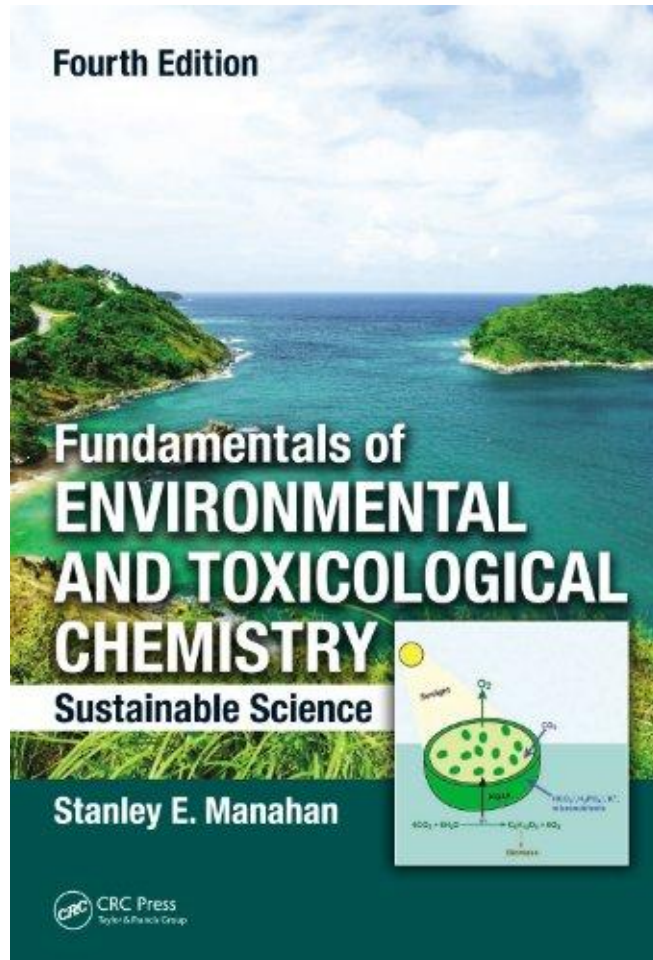
Course Content

Part 2 (Dr Abhijit): Aspects of transformations in atmosphere (microbial degradation of organics, environmental degradation of polymers, atmospheric lifetime, toxicity). Green chemistry and industrial Ecology.



Environmental Chemistry

Reference books by Stanley



Microbial degradation of organics & Environmental degradation of polymers



The major groups of microorganisms include bacteria, archaea, fungi, protozoa and algae

Microbial degradation of organics & Environmental degradation of polymers

Biodegradation is the breakdown of organic matter by microorganisms:

Organic matter is the carbon-rich material derived from living organisms and their decomposed remains, found in soil, water, and other natural environments. It plays a vital role in maintaining healthy ecosystems and fertile soils.

Bioremediation : use of either naturally occurring or deliberately introduced microorganisms to consume and break down environmental pollutants

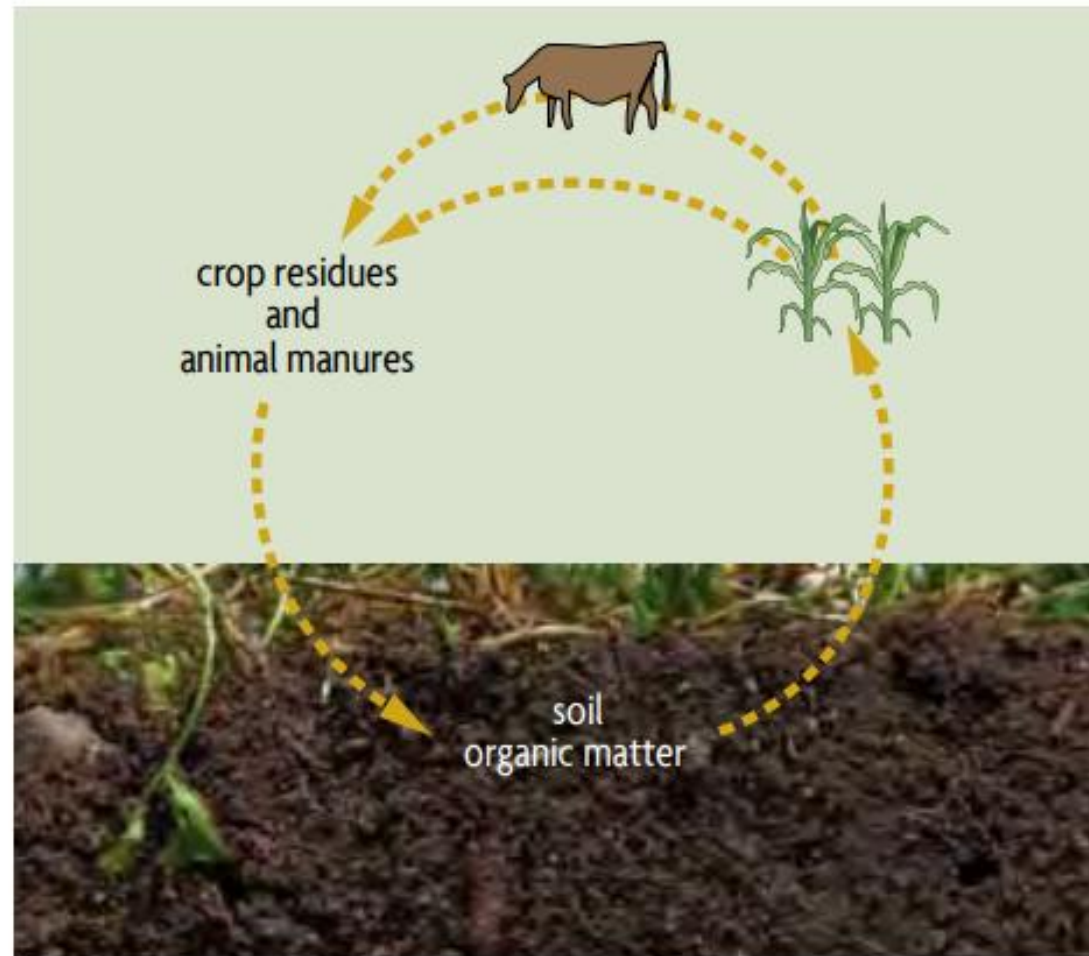
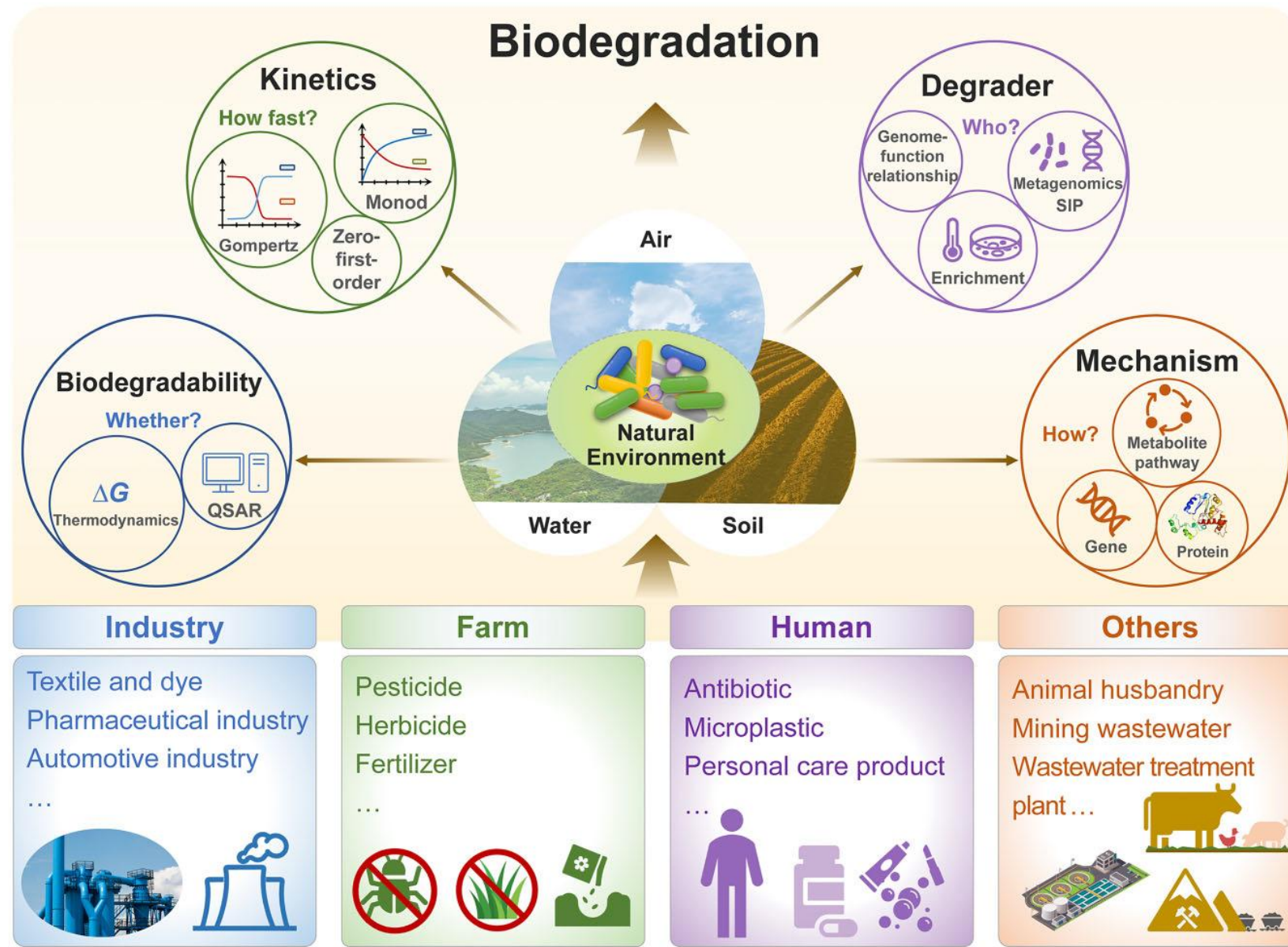


Figure 2.4. The cycle of plant nutrients.

Soil consists of four parts: solid mineral particles, water, air and organic matter. The particles are generally of sand, which consist of silicon, oxygen, aluminum, potassium, calcium, magnesium, phosphorus, potassium and other minor chemical elements

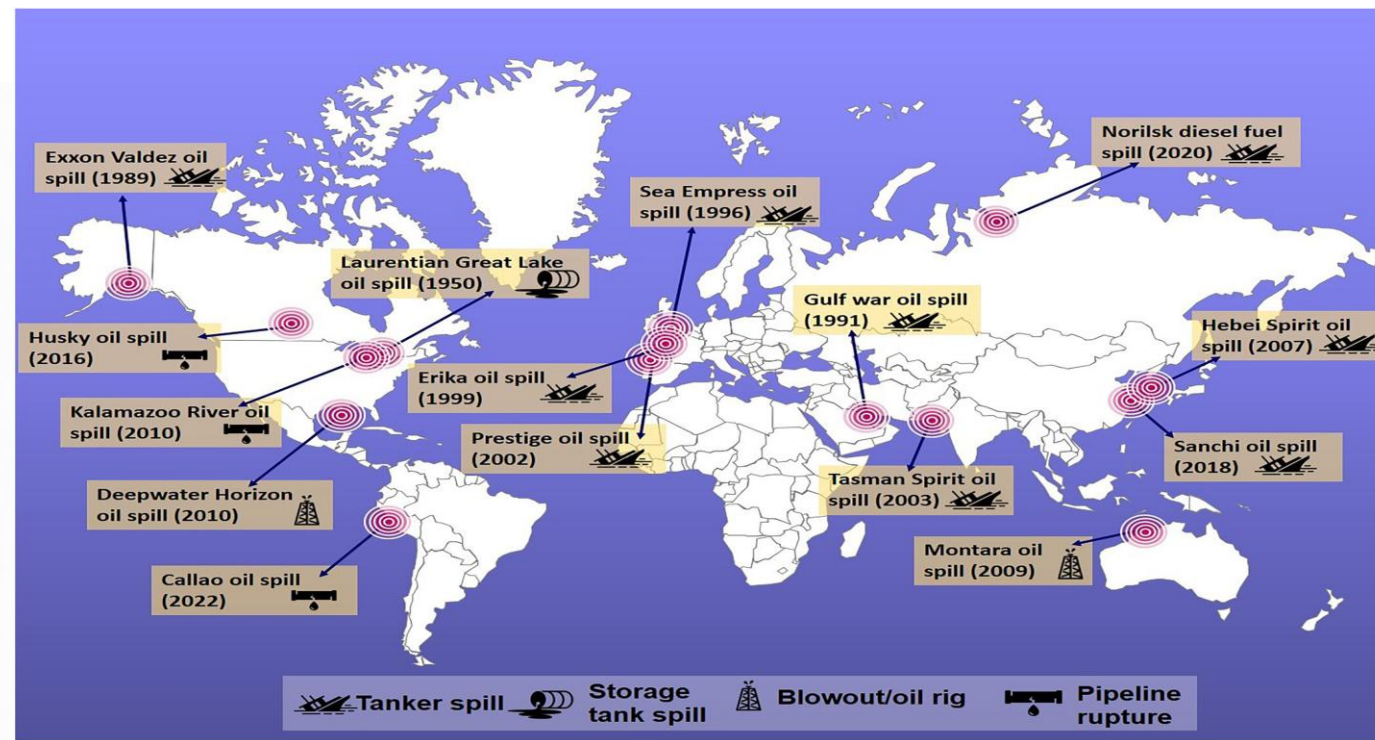


Basic Concept of Bioremediation

- Bioremediation uses living organisms, mainly plants and microorganisms, to degrade and detoxify environmental pollutants.
- Microorganisms such as bacteria, fungi, and algae convert it into CO₂, H₂O, microbial biomass or less-toxic organic compounds.
- Pollutants are degraded through microbial metabolic activities, where contaminants serve as energy and nutrient sources.
- Effective bioremediation requires favourable conditions, including appropriate pollutant type, solubility, bioavailability, and suitable environmental conditions for microbial growth.

OIL SPILL

Oil spills are among the most devastating environmental disasters, causing long-term damage to marine ecosystems, wildlife and coastal communities.



Oil slick in the ocean

Oil pollution spreads over vast areas, harming marine life and disrupting ecosystems.

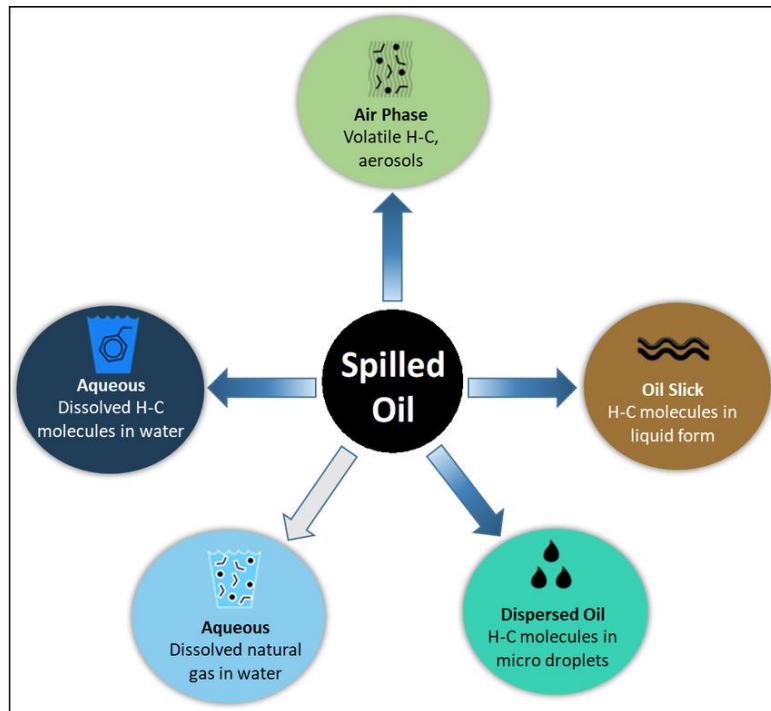
The Guardian



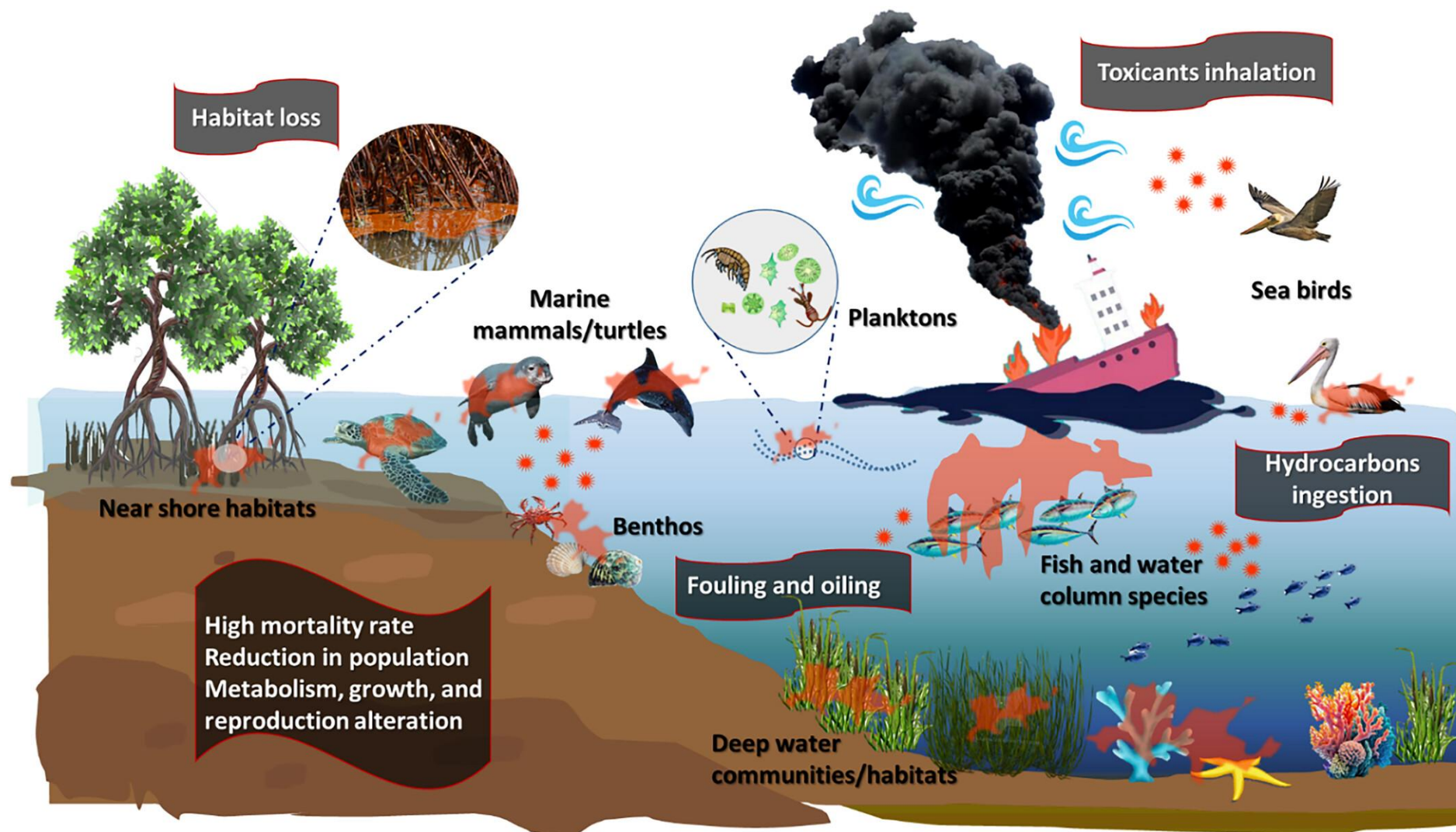
Wildlife affected by oil spill

Oil spills can be fatal to birds and other wildlife, destroying habitats and food sources.

- **Tanker Accidents**
- **Offshore Drilling Accidents**
- **Pipeline Leaks**
- **Equipment Failure**
- **Natural Disasters**



Schematic view of the impacts of spilled oil on aquatic ecosystems.



Remediation

In situ burning

In situ burning removes oil from the water surface by burning fresh oil under controlled conditions, achieving high cleanup efficiency but producing smoke and harmful air pollutants.

Mechanical remediation

Mechanical remediation uses physical tools such as booms, skimmers, vacuum trucks, and adsorbents to contain and remove oil from water without changing its chemical properties.

Chemical remediation

Chemical remediation employs dispersants, surfactants, and solidifiers to break down or solidify spilled oil for easier cleanup, although some chemicals may cause secondary environmental pollution.

Biological remediation

Biological remediation uses microorganisms such as bacteria and fungi to naturally degrade spilled oil, providing an environmentally friendly and cost-effective cleanup method.

Biodegradation of oil

Microbial oxidation, also known as biodegradation, is a process where microbes, primarily bacteria and fungi, break down the hydrocarbons in the oil and those derived from photo-oxidation into simpler compounds.

Biodegradation takes place at the oil–water interface with the presence of oxygen. Warmer climate is favorable for microbial activities,

Marine waters host a diverse array of microorganisms capable of utilizing hydrocarbons as an energy source, leading to the partial or complete degradation of oil into water-soluble compounds, and producing carbon dioxide and water.

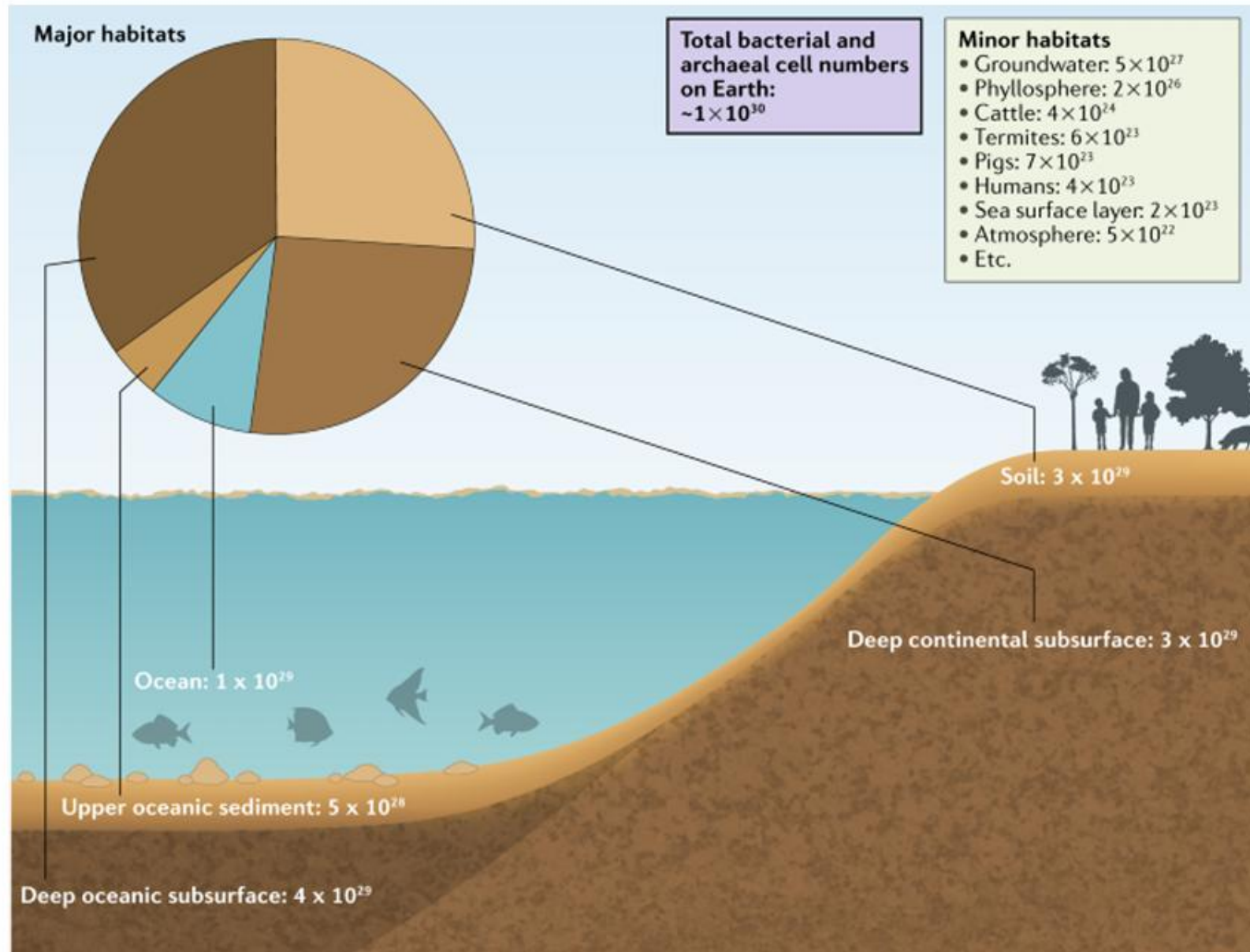


Microorganisms eat this Oil

Table 1 Microbes involved in degradation of petroleum hydrocarbons and their preferred degradation substrates

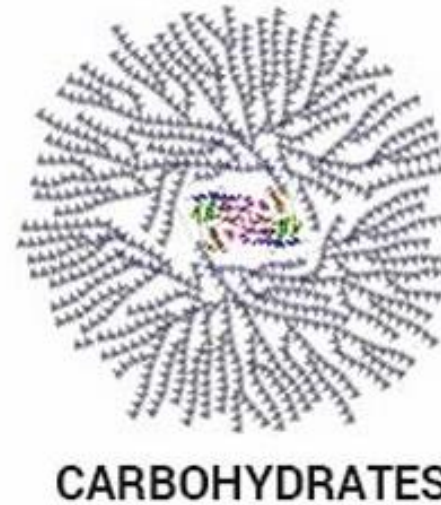
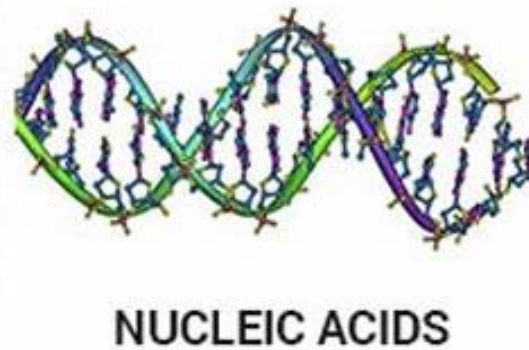
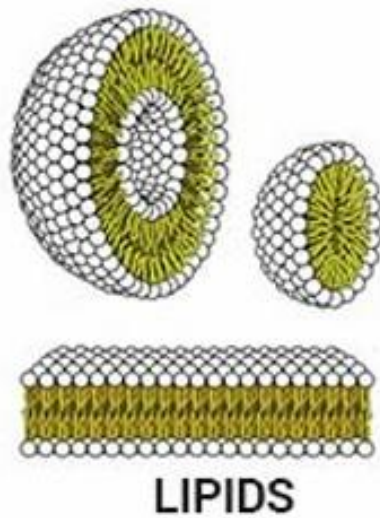
Petroleum hydrocarbon components	Bacterial species	Main degradation profile	Reference
Aliphatics	<i>Dietzia sp.</i>	n- alkanes (C ₆ -C ₄₀)	[59]
	<i>Pseudomonas sp.</i>	n-alkanes (C ₁₄ -C ₃₀)	[60]
	<i>Oleispira Antarctica</i>	n-alkanes (C ₁₀ -C ₁₈)	[61]
	<i>Rhodococcus ruber</i>	n-alkanes (C ₁₃ -C ₁₇)	[62]
	<i>Geobacillus thermodenitrifican</i>	n-alkanes (C ₁₅ -C ₃₆)	[63]
	<i>Rhodococcus sp.</i>	Cyclohexane	[64]
	<i>Alcanivorax sp.</i>	n-alkanes and branched alkanes	[65]
	<i>Gordonia sihwensis</i>	Branched and normal alkanes	[66]
Aromatics	<i>Achromobacter xylosoxidans</i>	Mono- /polyaromatics	[67]
	<i>Aeribacillus pallidus</i>	Mono- /polyaromatics	[68]
	<i>Mycobacterium cosmeticum</i>	Monoaromatics	[69]
	<i>Pseudomonas aeruginosa</i>	Monoaromatics	[70]
	<i>Cycloclasticus</i>	Polyaromatics	[71]
	<i>Neptunomonas naphthovorana</i>	Polyaromatics	[72]
	<i>Bacillus Licheniformis Bacillus Mojavensis</i>	Polyaromatics	[73]
Resins and asphaltenes	<i>Sphingomonas Sphingobium Novosphingobium</i>	Polyaromatics	[74]
	<i>Pseudomonas sp.</i>	Resins	[75]
	<i>Pseudomonas spp.</i>	Asphaltenes	[76]
	<i>Bacillus sp.</i>		
	<i>Citrobacter sp.</i>	Asphaltenes	[77]
	<i>Enterobacter sp. Staphylococcus sp. Lysinibacillus sp.</i>		
	<i>Bacillus sp.</i>		
	<i>Pseudomonas sp.</i>		

Total bacterial and archaeal cell numbers on Earth



Mechanism of Microbial Degradation of Organic Pollutants

Biochemistry and Toxicology



Inside the cell, microorganisms secrete enzymes that initiate degradation.

Common enzymes

- Oxygenases (monooxygenases and dioxygenases)
- Dehydrogenases
- Hydrolases
- Peroxidases
- Laccases (mainly in fungi)

These enzymes introduce oxygen or other reactive groups into the pollutant, making it more susceptible to further degradation.

Metabolic Conversion

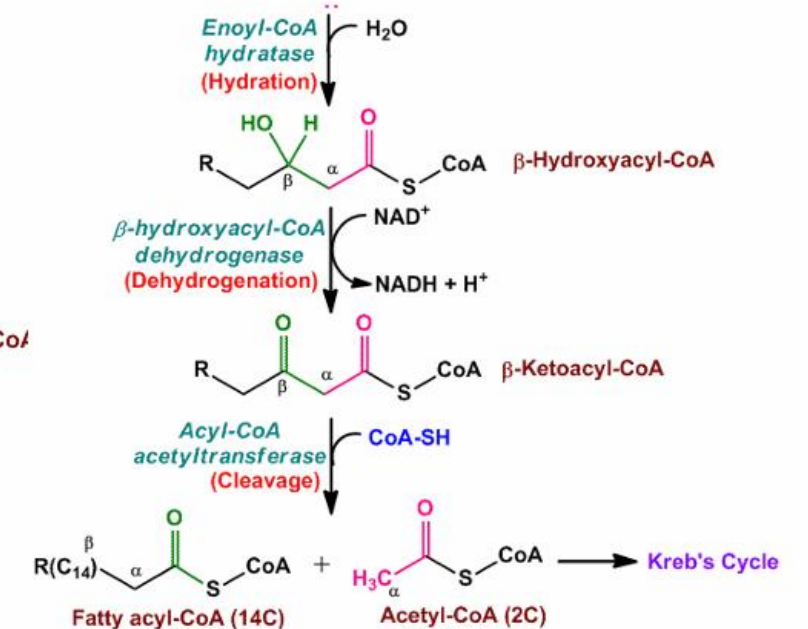
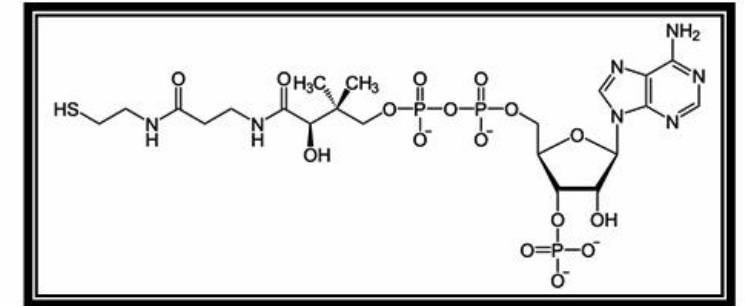
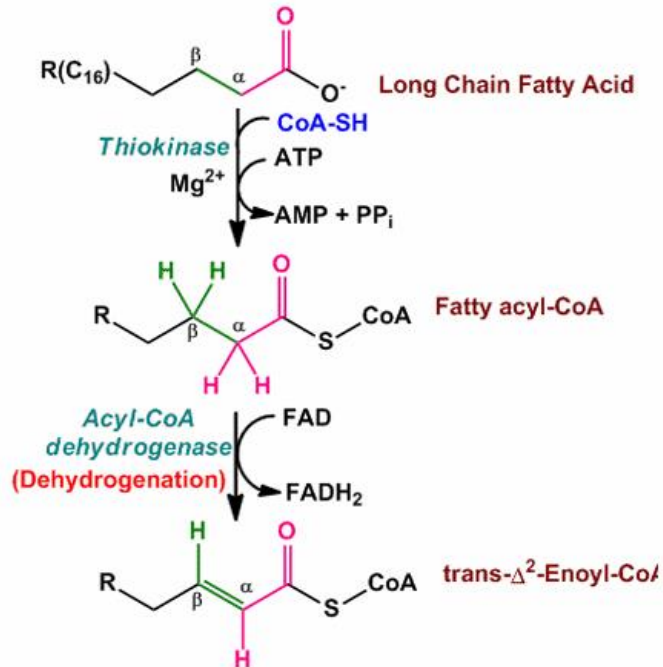
The intermediate compounds enter central metabolic pathways such as:

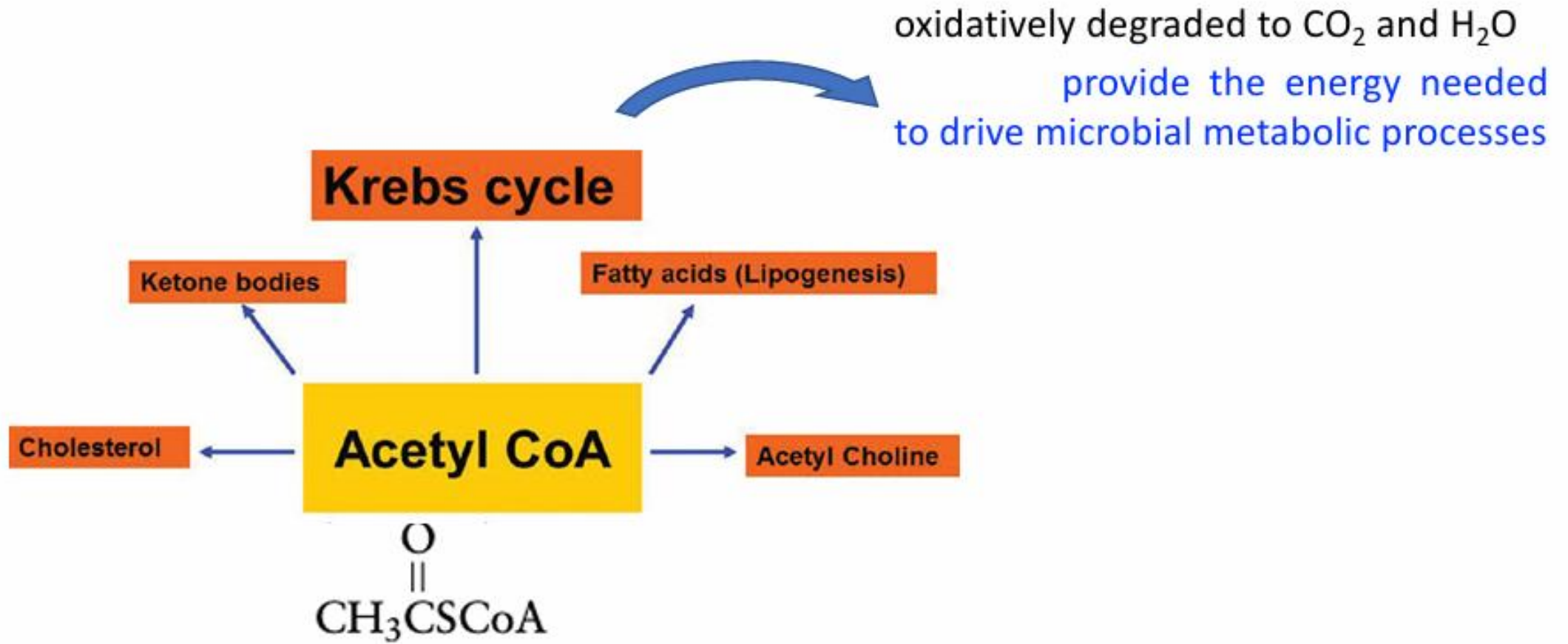
- **β -oxidation** (for fatty acids and hydrocarbons)
- **Citric acid (TCA) cycle**
- **Glycolysis**
- **Electron transport chain**

β-oxidation

To overcome the relative stability of the C-C bonds in a fatty acid, the carboxyl group at C-1 is activated by attachment to coenzyme A, which allows stepwise oxidation of the fatty acyl group at the C-3, (**β**-position) hence the name oxidation.

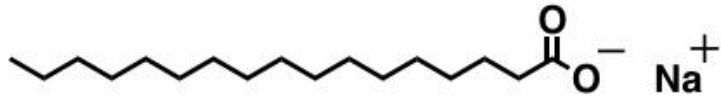
β- oxidation procedure



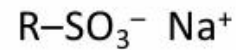


The rate of degradation by other environmental reagents such as sunlight, oxygen, or water is much slower than that of microbial degradation

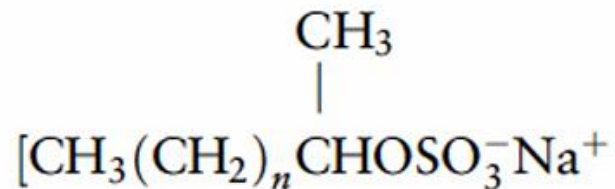
Soaps



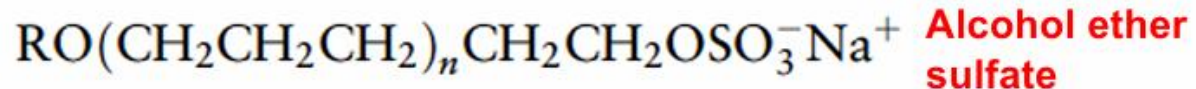
Synthetic surfactants



Alkyl sulfonates



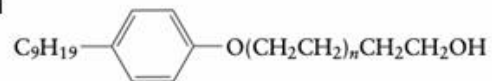
Alcohol sulfate



Alcohol ether sulfate

Non-ionic Alcohol ethoxylates $\text{RO}(\text{CH}_2\text{CH}_2)_n\text{CH}_2\text{CH}_2\text{OH}$

Alkylphenol ethoxylates $\text{R-(p-benzene)-O}(\text{CH}_2\text{CH}_2)_n\text{CH}_2\text{CH}_2\text{OH}$

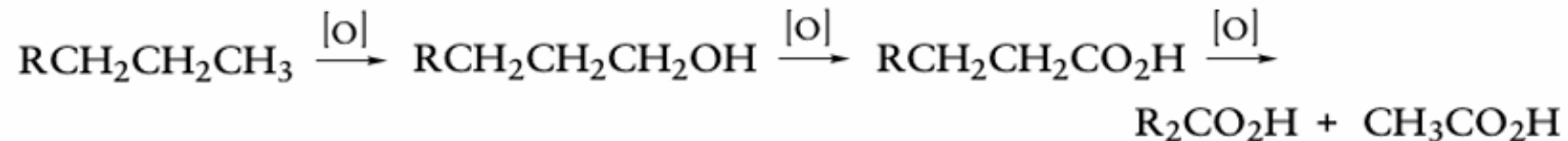


The microbial breakdown of detergents and petroleum

It requires the terminal oxidation of the hydrocarbon chain to a carboxylic acid followed by the β -oxidation procedure.

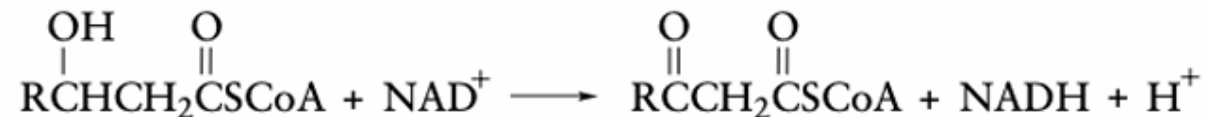
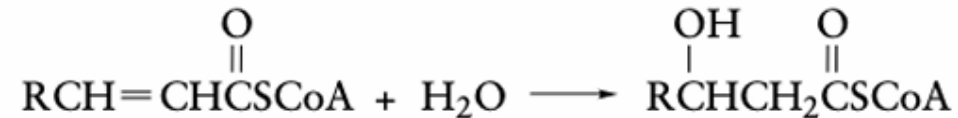
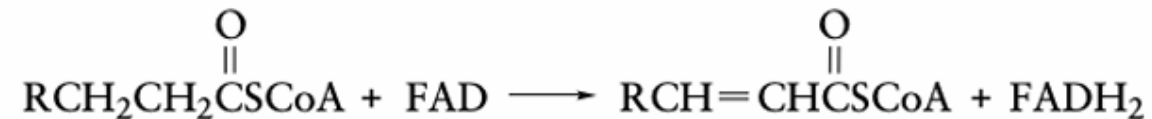
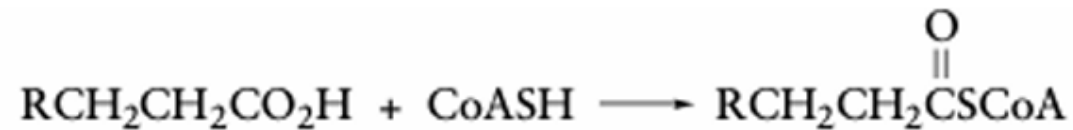
This terminal oxidation is a well-documented process:

Molecular oxygen is the oxidizing agent, and the oxygen is activated by **binding to the iron-containing enzyme cytochrome P-450 (function as Monooxygenases)**



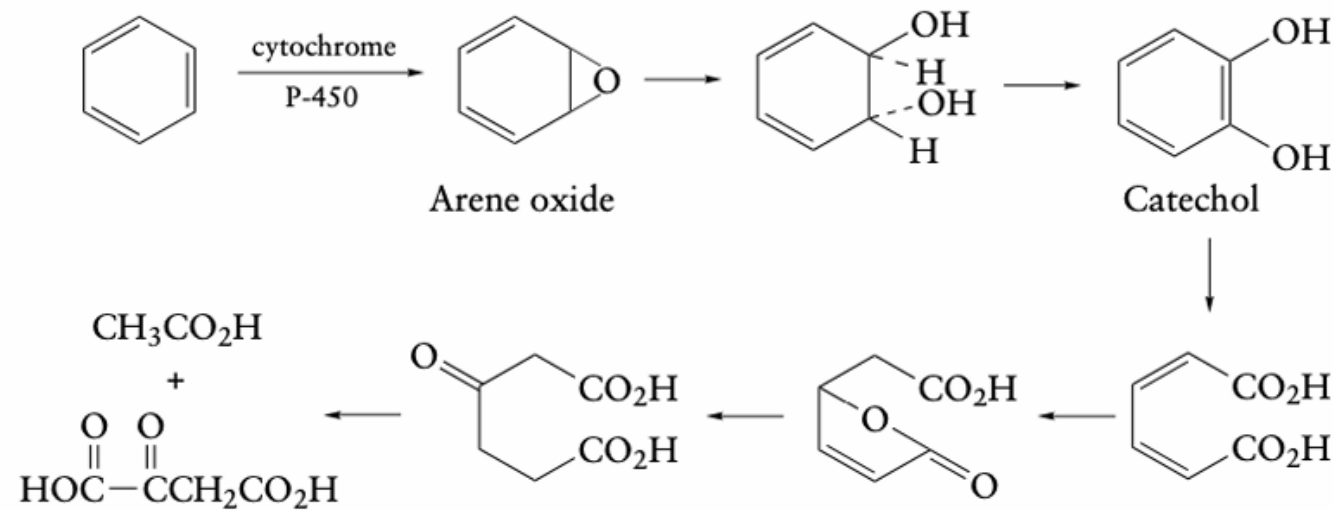
but are slow, especially with high molecular weight materials that do not readily pass through the microbial cell wall

- Fatty acids (including protonated soaps) occur widely in biological systems and are therefore readily degraded by microorganisms in enzyme-catalyzed reactions.
- Linear hydrocarbons and linear detergents are similar in structure to soaps, and these are degraded in the same or similar pathways.

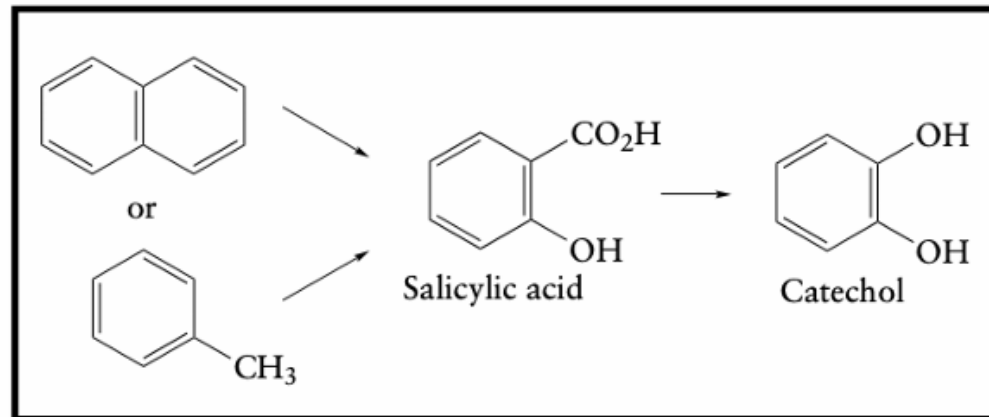


Oxidation of the cyclic hydrocarbons and aromatic compounds

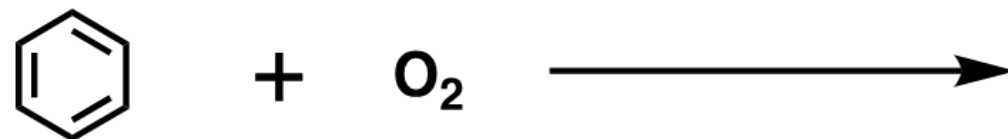
The microbial oxidation of the cyclic hydrocarbons and aromatic compounds present in detergents and petroleum is also catalysed by the enzyme cytochrome P-450 in microorganisms.



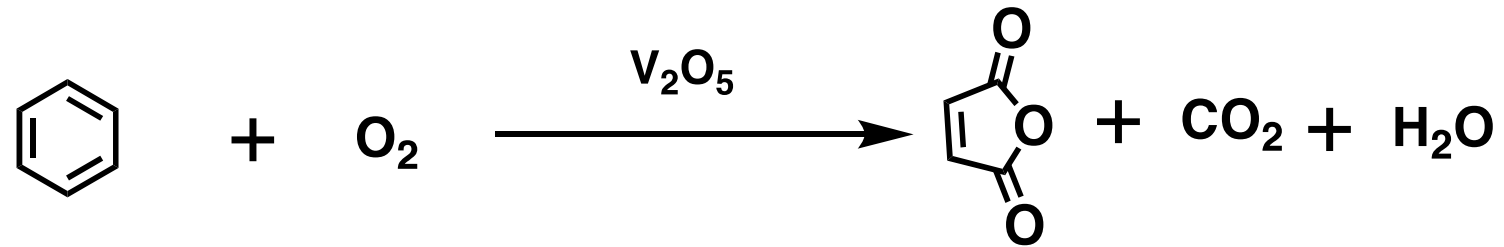
more highly condensed aromatics
proceeds via salicylic acid



Chemical Oxidation of Benzene



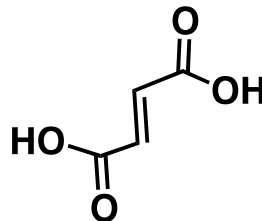
Chemical Oxidation of Benzene



Oxidation of benzene produce maleic anhydride

Benzene and air mixture is oxidized to maleic anhydride over the catalyst at 2-5 bar, 400-450⁰C,

Byproducts:



Fumaric acid

